

6.1 Software Engineering

Code / Terminal



Interface

- Repository
- Execution (bash)
- Unit tests

Feedback

- STRONG
(hard oracle)

Key Techniques

- Repository Grounding
- Workflow-Level Planning
- Feedback-Driven Repair

6.2 Information Seeking

Deep Research



Interface

- Web
- Search engines
- Documents

Feedback

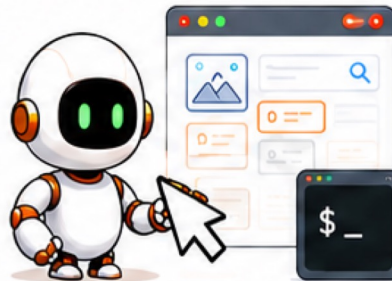
- WEAK
(evidence only)

Key Techniques

- Deep Search
- Wide Search
- Multimodal Grounding
- Research Synthesis

6.3 Computer Use

GUI / CUA



Interface

- Screenshots (pixels)
- DOM / UI state
- Shell / Keyboard

Feedback

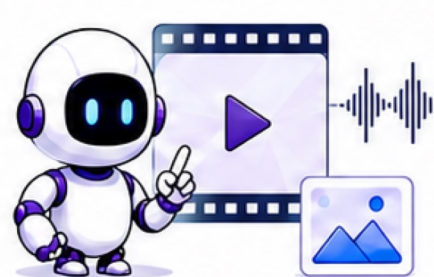
- PARTIAL
(delayed state)

Key Techniques

- Browser Agents
- Desktop GUI Agents
- Mobile Agents

6.4 Multimodal Agents

Long sensory streams



Interface

- Video streams
- Audio streams
- Images / frames

Feedback

- SPARSE
(implicit)

Key Techniques

- Multimodal Understanding
- Multimodal Generation
- Omnimodal Agency

6.5 General-purpose Agents

Open-ended tasks



Interface

- Many tools & APIs
- Many interfaces
- Humans & env

Feedback

- MIXED
(implicit & diverse)

Key Techniques

- Personal Assistants
- Embodied Agents and World Models
- Productive Agents

Bounded,
Verifiable Environment

Open,
Heterogeneous World